MULTIMEDIA COMMUNICATION

TELE SERVICES
INTERACTIVE SERVICES
DISTRIBUTED SERVICES
IMPLEMENTATION
APPLICATIONS

MEDIA COMMUNICATION AND TELE-SERVICES

Exchanging different multimedia content over a network.

Achieved through tele-services.

- based on AND make use of audio and video data.

Video conferencing, cooperative work, mailing, etc.

End users are located in different places, but can

- interact closely.
- operate on remote data and resources.

Delivery time is longer than processing time of local applications.

INTERACTIVE SERVICES

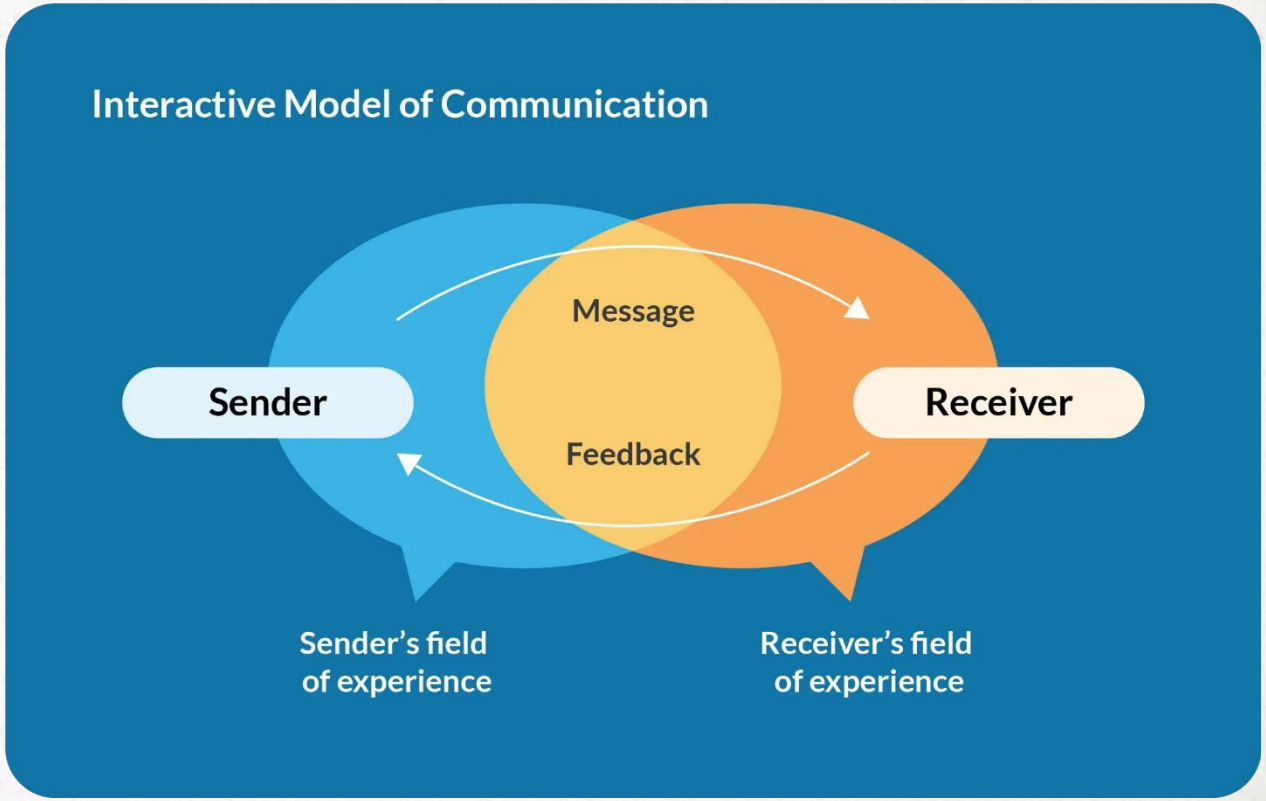
Exchange of control data between remote sites.

Synchronous: data arrives with a well-defined end-to-end delay.

- Video conferencing

Asynchronous: data arrives at any time.

- Mailing systems



Types of Interactive Services

- Conversational Services
- Messaging Services
- Retrieval Services
- Tele-Action Services
- Tele-Operation Services

	Conversational Service	Messaging Service	Retrieval Service	Tele-action Service	Tele-operation Service
Definition	Conversation between remotely located users.	Exchange of messages between sender and receiver, both humans.	Client is a human user, and server is a computer with database provision.	Reading/writing info from or to a remote location via a communication network.	Perform a task at a distance by manipulating a master that causes slave movement.
Example	Video Conferencing	Mailing Systems	Document Retrieval Systems	Banking Systems	Tele-Robotics System
Features	Synchronous	Asynchronous	Client to Server – Async Server to Client – Sync/Async	Data Collection + Action Initiation	A telemanipulator controlled by a human operator + robots controlled by the telemanipulator
	Minimal end-to-end delay	Fast delivery time is not necessary	Strict reliability of data, good delivery time		

DISTRIBUTION SERVICES

Distribution Services

- Services for distribution of information.
- One-way communication – broadcast source to remote clients.

Pay-per-view

- Cable subscribers ordering movies and programs.
- No control over what movie is shown.

Near VoD

- Many channels broadcast the same program, but with a temporal variation.
- Viewer can simulate forward/reverse functions by switching channels.

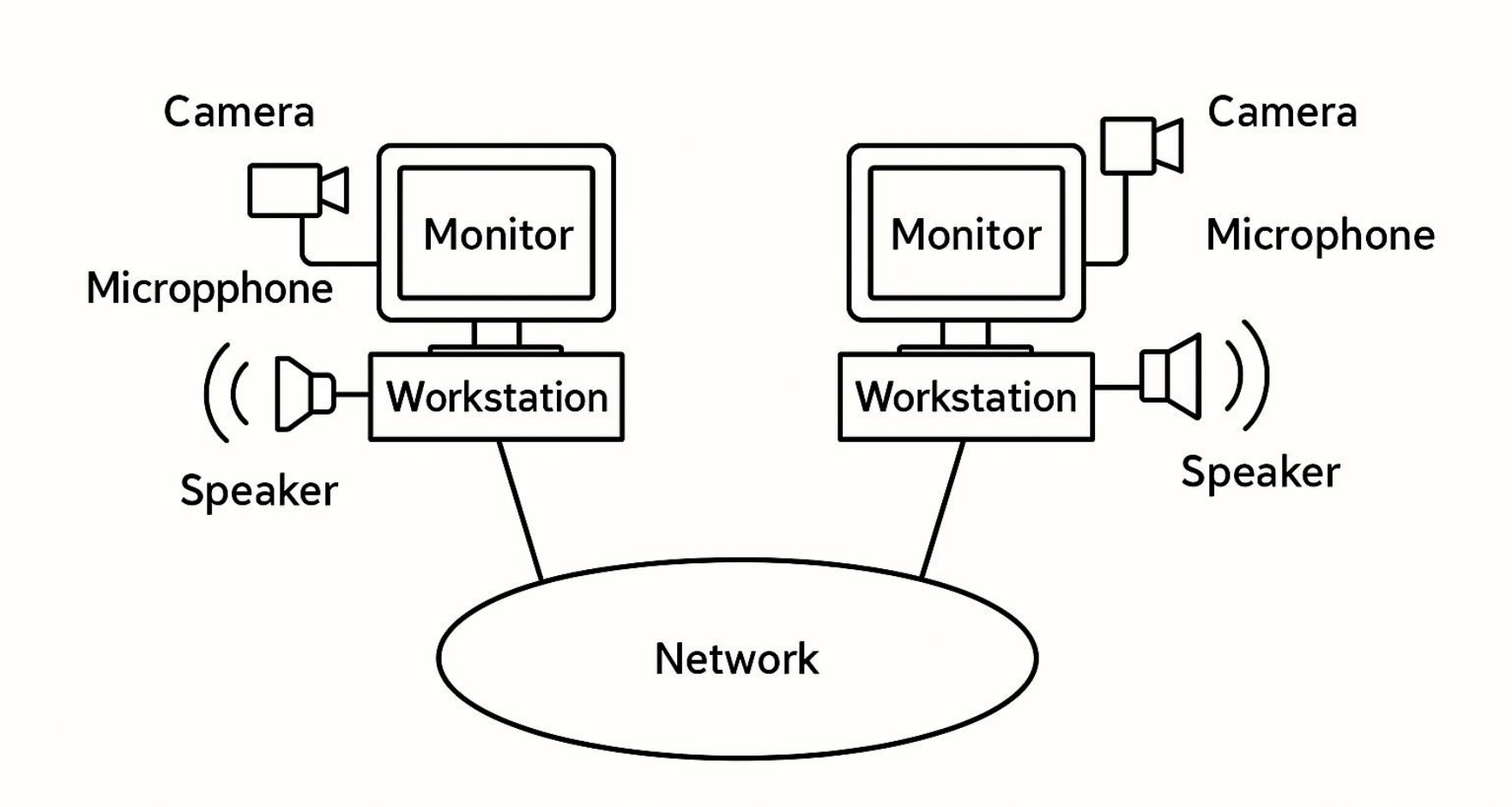
True VoD

- Complete freedom to temporally alter viewing.
- Interactive TV/Cyber Vision: user is involved in the content.

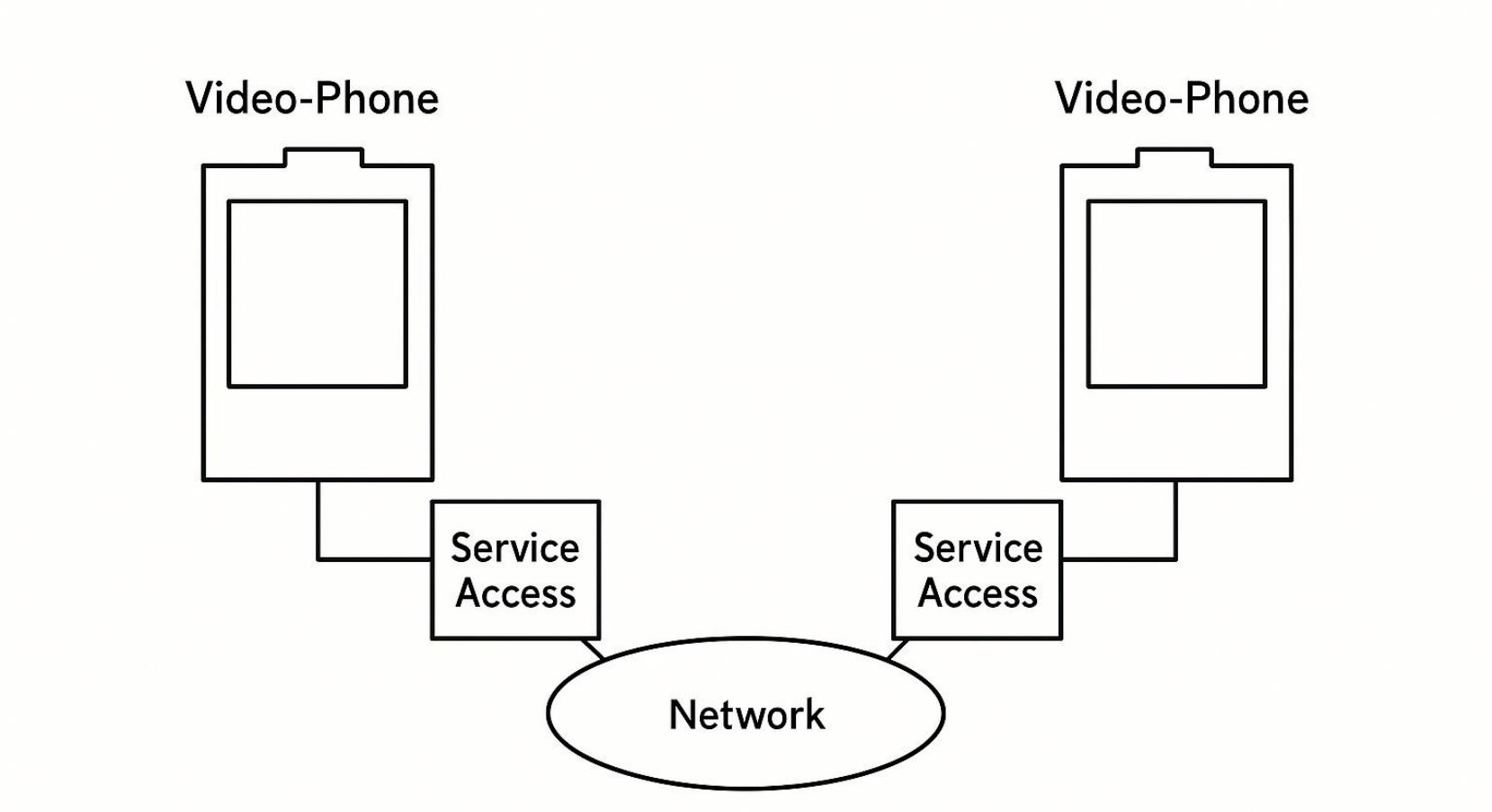
IMPLEMENTATION

Conversational Services	Video Conferencing
	Video-Telephony
	Computer Supported Cooperative Work
Messaging Services	MIME
	Multimedia Mail Tele-Service
Retrieval Services	Video Server
	World Wide Web
	Internet Gopher
Tele-action Services	Utility Resource Management
	Alarm and Surveillance
Tele-operation Services	Tele-Robotics System

VIDEO CONFERENCING



VIDEO TELEPHONY



COMPUTER SUPPORTED COOPERATIVE WORK

Cooperative work done in an infrastructure of networked stations and PCs.

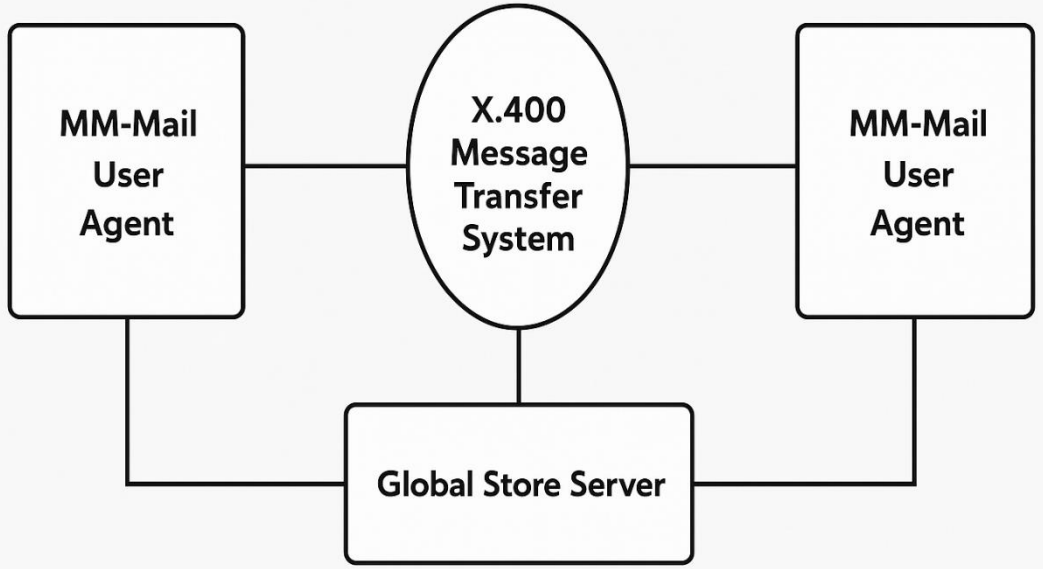
Allows several people at different locations to work on the same data.

Can be synchronous or asynchronous.

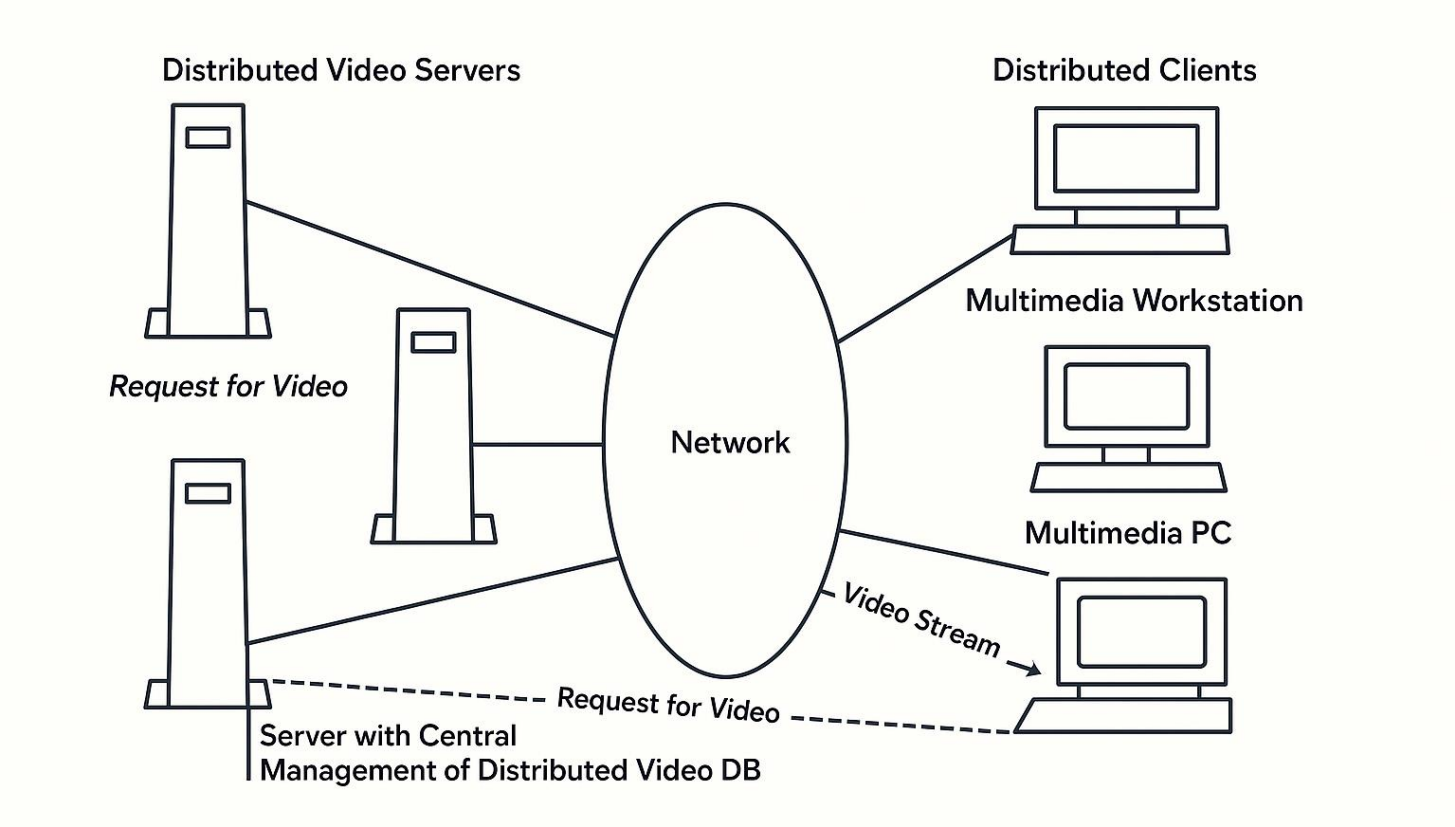
Systems that support such CSCW are called groupware.

BERKOM MULTIMEDIA MAIL TELE-SERVICE

What	Tele-service for transmission of multimedia content over mail.
X.400	MTS is based on this, a messaging standard for the OSI model.
Messages	Can be from MM-Mail UA or GSS.
Multimedia	Store and forward mechanism; uses references for larger data.



VIDEO SERVER



INTERNET GOPHER

What

- Communication protocol for distributing, searching, and retrieving documents
- Predecessor of WWW, fell into disfavour.

Who

- Invented by the University of Minnesota.
- Released and quickly become established in Mid-1991, when WWW was in its infancy.

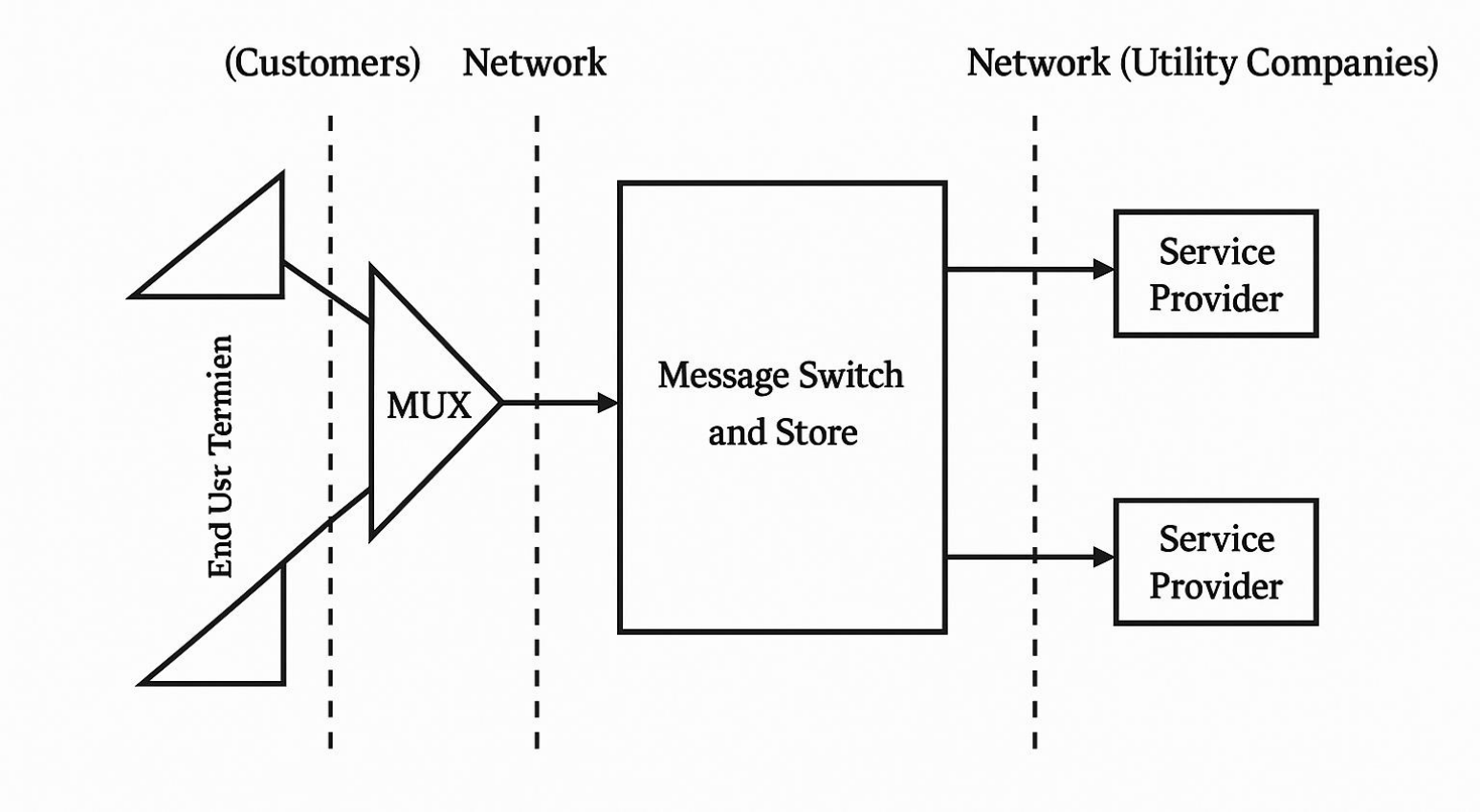
Central Goals

- File-like hierarchical management, and a simple syntax.
- A system that can be created quickly and inexpensively.

Gopher Objects

- Text
- Sound
- Images
- Files
- MIME information
- HTML documents

UTILITY RESOURCE MANAGEMENT



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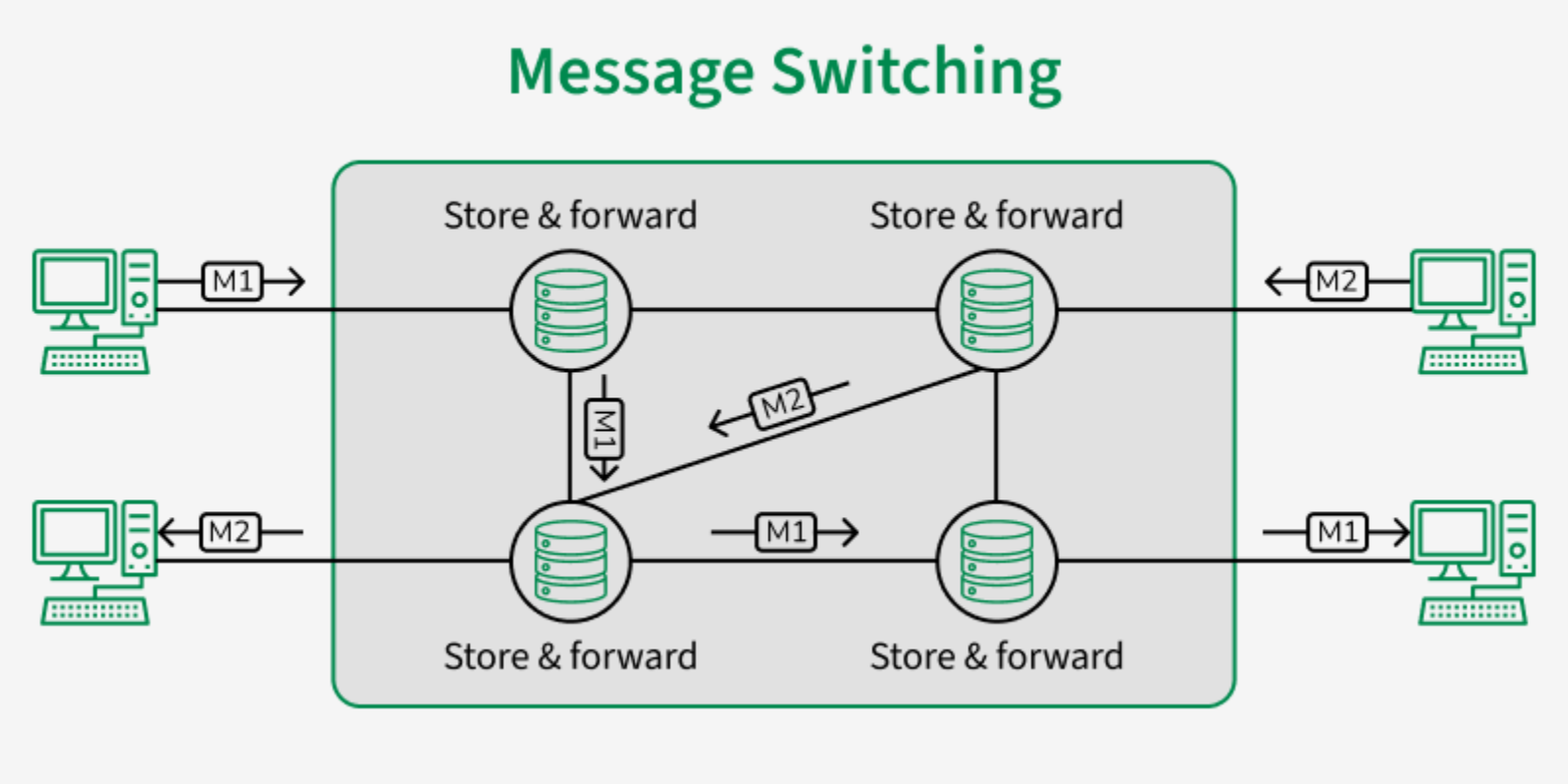
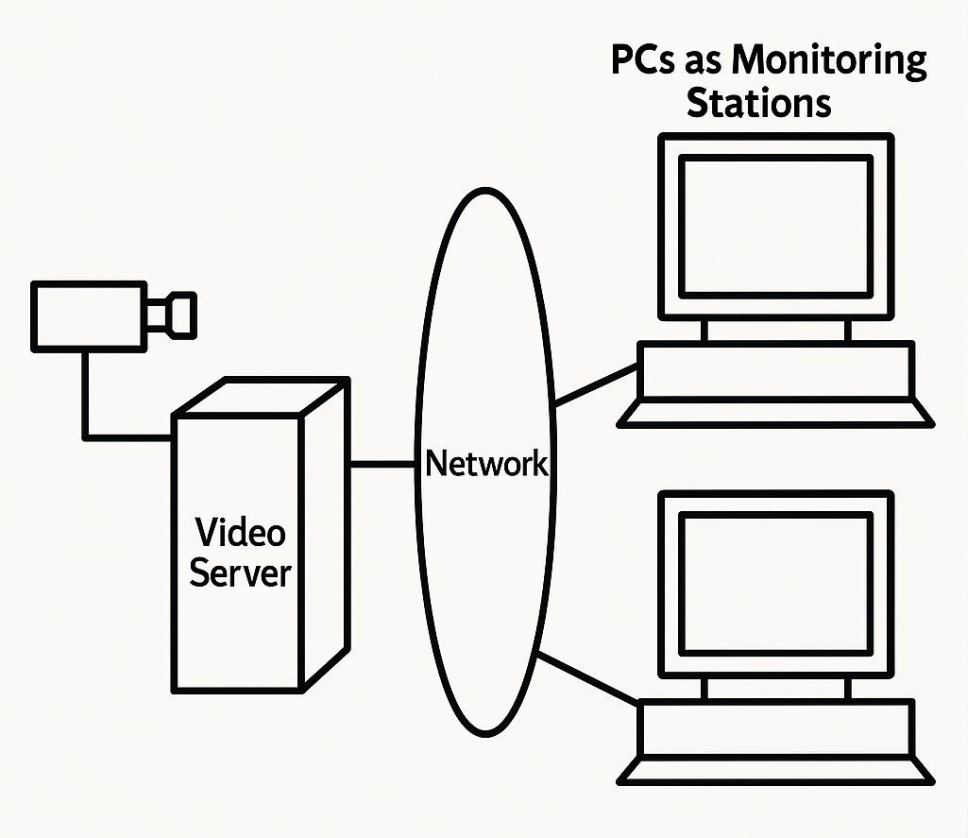
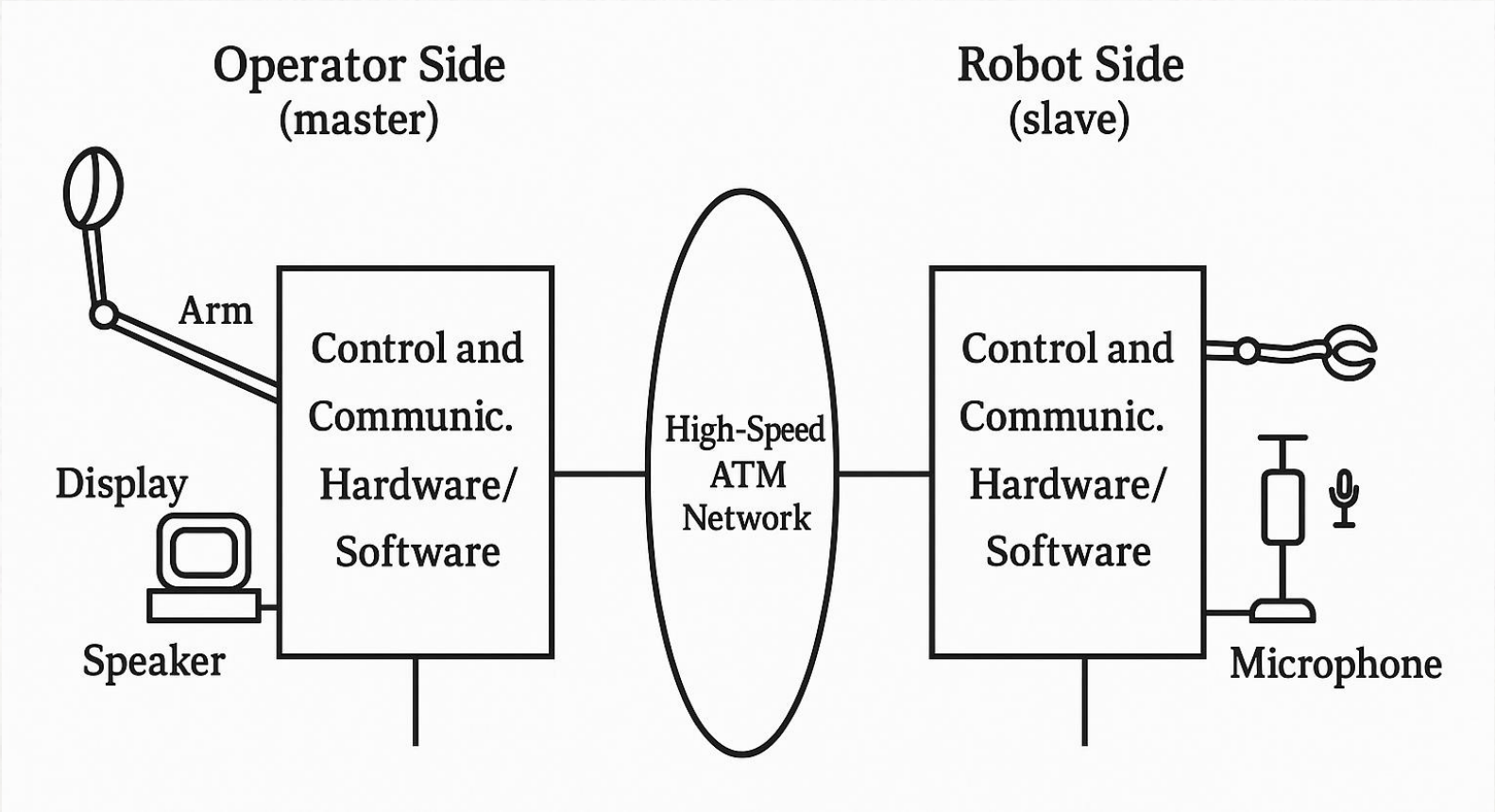


ALARM AND SURVEILLANCE SERVICE



TELE-ROBOTICS SYSTEM



APPLICATIONS

Remote Learning

Multimedia remote tutoring services.

Tele-teaching
Real-time tele-assistance
Real-time multimedia conferencing
Interactive distance learning

BETEL	**B**roadband **E**xchange for **T**rans-**E**uropean **L**inks
BETEUS	**B**roadband **E**xchange for **T**rans-**E**uropean **US**age

APPLICATIONS

Tele-Office

Work from Home

PC
Printer
Phone
Video-Wall
Fax

APPLICATIONS

Tele-Collaboration

Media Space: a work environment that basically enables tele-collaboration.

Network
Hardware
Software

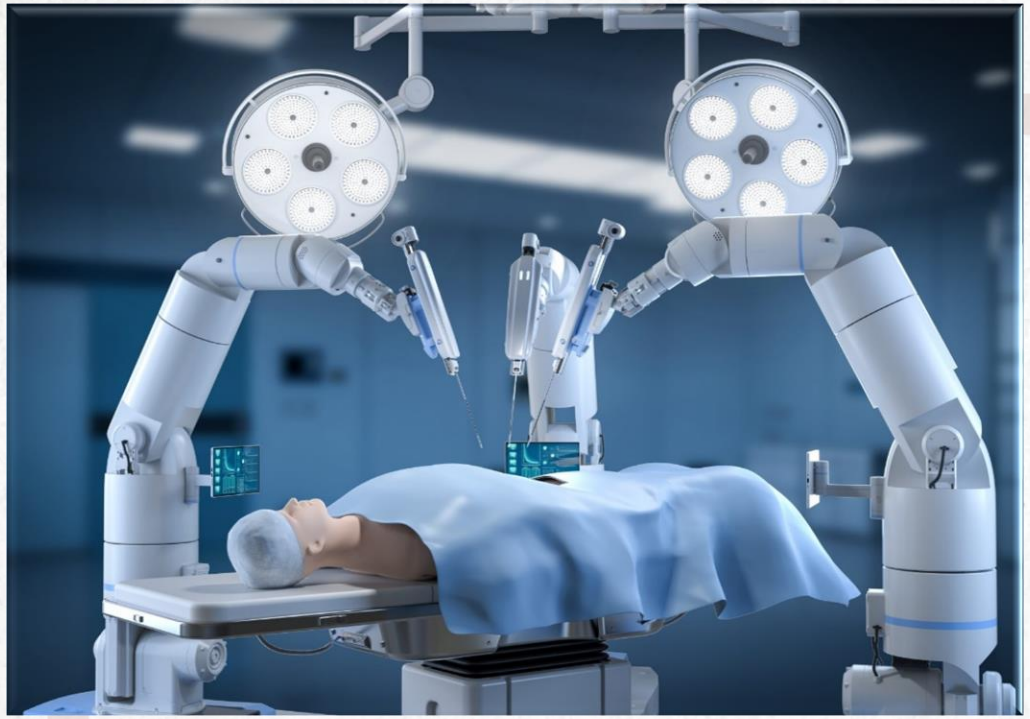
APPLICATIONS

Tele-Medicine

Addressing elderly, sick, or disabled people who can't leave their homes or are advised not to.

Consult the doctor, get medical and health information.

APPLICATIONS



Tele-Surgery

Using remote controlled devices to perform surgery.

To ensure better precision.

Or to consult a surgical specialist from a remote location for crucial operations.

THANK YOU!

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